

Hybrid Connected Kitchens: Phygital Social XR for Domestic Communication and Collaboration

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Abstract

Extended Reality (XR) technologies are increasingly used to support communication, collaboration, and social connection across distance. While social XR research has primarily focused on workplaces, education, and entertainment, far less is known about how hybrid physical–virtual communication might unfold within everyday domestic environments—particularly kitchens, where cooking, coordination, and care shape social life. This paper investigates how design practitioners envision social interaction and collaboration if kitchens become hybrid connected spaces augmented through phygital artifacts. We conducted a designer-focused evaluation using three speculative prototypes—an ambient cooking companion, a synchronous communication portal, and an asynchronous three-dimensional messaging system—embedded within an immersive VR kitchen environment. Through situated tasks, think-aloud VR sessions, semi-structured interviews, and a structured design-evaluation survey, we examined how designers conceptualize communicative and relational possibilities, how they imagine practices emerging around hybrid kitchen artifacts, and what tensions arise when XR enters domestic life. Findings show how connected kitchens can function as phygital communication hubs that support synchronous, ambient, and asynchronous social presence; how designers balance opportunities for togetherness with concerns around privacy, household rhythms, and inclusivity; and how immersive speculative prototyping enables early-stage reasoning about domestic XR futures. We conclude by outlining design implications for socially meaningful, equitable, and ethically grounded connected kitchen ecologies and identify directions for future work on long-term domestic appropriation, cross-household interaction, and broader spatial ecologies of hybrid home life.

Keywords: Social XR, Domestic XR, Connected kitchens, Phygital communication, Hybrid interaction, Speculative design, Collaboration

1 Introduction

As wearable computing, generative AI, and connected devices converge with Extended Reality (XR) technologies—including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), they are reshaping how people connect, collaborate, and sustain social relationships across distance. By blending digital content with physical environments or immersing users in fully virtual spaces, state-of-the-art XR systems promise alternative forms of communication enriched by embodied cues, shared spatial attention, and immersive social presence [1–3]. Yet as these technologies evolve, concerns remain about whether they can genuinely support meaningful social interaction, or whether they risk reproducing the limitations of screen-based media in three dimensions. While research on social XR has grown rapidly, it has predominantly examined work-oriented collaboration, education, entertainment, and user behavior in large-scale social platforms [2]. Far less attention has been paid to domestic environments, where communication and coordination intertwine with routines of care, intimacy, and shared living. Kitchens are particularly socially dense domestic settings: they are sites of teaching and learning, hosting, improvisation, negotiation, and emotional connection. In many families, cooking together or sharing food is a cornerstone of social life, one that increasingly extends to remote encounters as people cook together over the phone, video chat or send photos and messages about meals.

HCI research has begun exploring technologically mediated cooking and dining. Telematic dinner parties use multimodal setups to create playful, performative forms of togetherness [4], while virtual cook-along experiences using videoconferencing illustrate the opportunities and challenges of remote collaboration around food [5]. Yet such systems largely remain tied to flat screens and limited camera views, offering only partial support for the embodied, spatial, and sensory cues that shape cooking as a social practice. In parallel, AR and MR work has examined situated cooking assistance. *Shadow Cooking* demonstrated how spatial projection of instructions onto utensils and ingredients can enable fluid, embodied task performance [6]; inter-generational AR cooking tools show how overlays and quantity visualizations can reduce confusion and strengthen emotional connection between remote family members [7]. More recent work such as “Cookie” [8] highlights how multimodal smart kitchen assistants can support cooking while preserving user autonomy, balancing proactive system support with human agency. However, these systems primarily emphasize instruction, safety, and task performance rather than broader questions of social presence, asynchronous communication, ambient connection, or long-term household integration.

Beyond food contexts, domestic HCI research has explored how networked and ambient technologies can shape family life and co-presence. Situated displays that visualize physiological data support coordination and reflection in households with ADHD children [9]. Research-through-design explorations of passive co-presence—such as the *There Chair* and *Fragrance Frame*—illustrate how subtle tactile or olfactory cues support emotional connection without demanding attention [10]. Moreover, technology-probe studies of AR in the home show that users value AR not only for utility but also for its capacity to support social activity, awareness, and playful shared experiences in domestic life [11]. Collectively, these works suggest a design space of

phygital domestic experiences in which ambient, spatial, and embodied cues create a sense of “being together apart” that differs from traditional communication methods.

Additional research in smart homes and hybrid-reality systems highlights the value of coupling physical and virtual spaces. Hybrid smart home prototypes [12] and methodological platforms like AR-in-VR simulators [13] show how immersive environments can enable early-stage evaluation of domestic technologies. Recent work on VR-based kitchen prototyping further demonstrates how designers use immersive scenes to reflect on appliance interactions, workflows, and user experience in context-rich domestic environments [14]. Yet across these domains, the question of how people—especially designers—conceptualize *social* interaction, collaboration, and relational practices in hybrid kitchens remains largely unexplored. Despite growing interest in social XR, phygital interfaces, and domestic computing, we still lack an in-depth understanding of how designers envision the communicative and emotional possibilities of hybrid connected kitchens.

While prior smart-home and kitchen XR research has largely focused on task assistance, safety, interaction techniques, or procedural guidance, much less is known about how future hybrid kitchen systems might mediate social presence, everyday communication, and the subtle rhythms of domestic life. Existing systems tend to treat the kitchen as a technical workspace rather than a socially charged environment where routines of care, coordination, and shared meaning unfold. Our goal is therefore not to extend smart-kitchen automation, but to examine how *hybrid spatial communication* tools may reshape relational and communicative practices in domestic settings. To investigate these questions, we developed three speculative phygital artifacts; an ambient cooking companion (ambient channel), a synchronous communication portal (real-time channel), and an asynchronous 3D messaging system (temporal channel), embedded within an immersive VR kitchen environment. Together, these artifacts sketch a three-channel communication model for hybrid connected kitchens, representing different temporal and sensory modalities of domestic communication: ambient, synchronous, and asynchronous. To address this gap, we investigate: **(RQ1)** how professional designers interpret and appropriate speculative phygital artifacts within an immersive VR-connected kitchen as supports for synchronous, ambient, and asynchronous social interaction; and **(RQ2)** what opportunities and concerns they identify regarding social presence, household rhythms, inclusivity, privacy, and the broader socio-emotional dynamics of XR-enabled domestic ecologies.

To explore these questions, we conducted a designer-focused evaluation study featuring three speculative phygital artifacts—an ambient cooking companion, a synchronous communication portal, and an asynchronous three-dimensional messaging system—embedded within an immersive VR kitchen environment. Through situated VR tasks, think-aloud walkthroughs, semi-structured interviews, and a structured design-evaluation survey, we analyze how designers imagine everyday domestic practices emerging around these artifacts, where they see value and potential, and where tensions and ethical considerations surface. We focus intentionally on professional designers rather than end-users. Designers in home-appliance contexts act as *proxy futurists*—*experts* who routinely anticipate emerging technologies, interpret future domestic scenarios, and translate speculative ideas into product trajectories. Their

conceptualisations reveal early signals about how hybrid connected kitchens may be framed, appropriated, and problematised before such systems ever enter people’s homes. Studying designers therefore provides insight into the *future-making practices* that shape domestic XR long before widespread adoption.

Our investigation contributes a conceptual, empirical, and methodological foundation for understanding hybrid connected kitchens as socially meaningful XR environments. We advance the notion of kitchens as *phygital communication hubs*; provide insights into how designers reason about social presence, co-presence, ambient connection, accessibility, and privacy in domestic XR; and demonstrate how speculative VR-based prototyping can serve as an effective design-research method for probing future domestic technologies.

2 Background

Our work sits at the intersection of domestic XR, phygital interfaces, and social presence. We review how XR has been explored in kitchens, dining, and domestic contexts as sites of collaboration; how phygital and ambient interfaces connect physical artifacts, sensing infrastructures, and hybrid representations; how social presence and co-presence shape interaction in XR; and how inclusivity, accessibility, and ethics inform emerging domestic hybrid environments.

2.1 Social XR in Domestic Life and Connected Kitchens

XR has been widely explored for cooking guidance, food experiences, and domestic social encounters. Early projection-based systems such as Shadow Cooking coupled instructions with physical action [6], while recent reviews show how VR/AR support food preparation, dining, and affective manipulation [15]. Kitchen-focused XR further includes hands-free collaborative cooking systems like HoloCook [16], AR guidance tools that use multimodal prompts and context-aware overlays [17], and AR-based appliance support leveraging virtual assistants and 3D overlays [18].

Cooking has also become a site for empathy and sensory experimentation. *Cooking in the Dark* uses spatial audio, passive haptics, and voice interaction to simulate blindness and foreground accessibility [19]. VR co-cooking platforms explore social cohesion through synchronized cooking and shared gestural interaction [20]. Beyond kitchens, remote dining systems like the Telematic Dinner Party [4] and Mixed Dining for older adults [21] highlight hybrid togetherness around meals. Creative and design-oriented food XR extends these ideas. CakeVR supports remote cake co-design through shared 3D artifacts [22, 23], while KitchXR positions VR as a context-rich environment for understanding kitchen workflows and appliance use [14]. AR-enhanced intergenerational cooking reduces confusion and strengthens emotional connection between dispersed relatives [24]. More recent smart-kitchen work such as “Cookie” [8] explores multimodal assisted cooking that preserves human agency, balancing proactive system behavior with user control through Wizard-of-Oz prototyping. Virtual cook-along studies map the design space of remote cooking across planning, enabling, co-presence, and food interaction [5].

Existing work establishes kitchens and dining rooms as promising XR sites for guidance, empathy, co-creation, and remote togetherness [5, 6, 14, 15, 22, 24]. Yet most systems prioritize task assistance, entertainment, or single-purpose interactions, treating XR as a layer rather than as part of a broader hybrid communication ecology. We instead frame the *connected kitchen* as a social XR hub where phygital artifacts and an immersive VR prototype probe how domestic communication and cohesion might shift when physical and virtual layers become tightly interwoven.

2.2 Phygital Domestic Interfaces: Ambient Displays, Smart Homes, and Situated Guidance

Phygital and ambient domestic interfaces investigate how physical environments can host hybrid representations and subtle communication cues. Early ambient displays like Hubscape visualized home sensor data across surfaces [25], and implicit-interaction systems adapted content through proximity or identity sensing [26]. Hybrid smart homes use immersive models to evaluate context-aware services [27], while IoT Avatars link physical devices with manipulable mixed-reality representations [28]. Ambient artifacts increasingly support reflection and social awareness. Situated displays for ADHD co-regulation [9], family negotiation tools like Schlusslicht [29], and workspace awareness systems such as LiteCo [30] illustrate how subtle cues and continuous feedback shape routines. Always-on in-home energy displays reveal how ambient representations influence everyday decision-making [31]. Multi-sensory systems like LiveNature integrate visuals, audio, and tactile cues to support calm engagement for people with dementia [32]. Passive co-presence systems—There Chair, Fragrance Frame—highlight how peripheral tactile or olfactory traces sustain distant connection [10, 33]. Technology-probe studies of domestic AR reveal that users value augmented information not only for practical support but also for enhancing social activity, awareness, and playful shared experiences in the home [11]. This suggests that domestic mixed-reality systems must account for household rhythms, interpersonal dynamics, and the expressive dimensions of everyday communication.

Phygital smart-home control expands interaction through tangible and spatial interfaces. Everyday objects used as TUIs demonstrate how physical metaphors shape control literacy [34]. ARKitchen combines magnetic markers and iron workspaces for tangible AR cabinet authoring [35, 36]. MR interfaces explore hierarchical UI frameworks and near-object interactions [37], while comparisons of AR, MR, and VR highlight trade-offs in appliance control [38]. Systems like HoloHome integrate XR with Ambient Assisted Living for older adults [39], and AR-in-VR toolboxes enable prototyping domestic AR concepts under ecologically valid conditions [13]. Phygital domestic systems—ambient displays, IoT avatars, tangible controllers—show how hybrid cues can be woven into physical environments [25, 27, 28, 35, 38, 39]. Yet few explicitly target kitchens as sites of *social* communication or examine how such artifacts are experienced in synchronous social XR environments. We address this gap by embedding speculative phygital artifacts in a VR-connected kitchen and treating them as communicative media for awareness, togetherness, and domestic coordination.

2.3 Social Presence, Co-presence, and Collaboration in XR

Social presence and co-presence provide key lenses for understanding social interaction in XR. Foundational work highlights the illusion of non-mediation and sense of community as central to multi-user VR [1]. Reviews emphasize how visual feedback, embodied interaction, and motion fidelity shape emotional involvement and collaboration [3, 40]. Empirical studies examine how visual and embodied cues influence coordination. Avatar visualisation affects attention and interaction quality [41]. Behaviourally realistic bidirectional gaze reduces coordination overhead [42]. Eye tracking in XR maps gaze as input, communicative cue, and shared attention signal [43]. Movement fidelity enhances presence and influences interpersonal distance [44]. Avatar realism can strengthen trust and social presence [45, 46], though effects vary across contexts [47, 48].

Collaboration studies compare symmetric and asymmetric media configurations, showing benefits for hybrid PC–VR collaboration [49]. Co-located VR game collaborations reveal how roles, task structures, and communication channels shape collective efficacy [50]. In remote XR collaboration, 3D reconstruction views often outperform 2D or 360° video [51]. SocialVR platforms using volumetric reconstruction support emotion perception and social presence [52]. Multimodal cues—vibrotactile, haptic, physiological—further modulate connectedness and coordination [53–55]. Several studies complicate simple presence–performance relationships. AR-based remote guidance shows social presence is not always predictive of collaboration quality [56]. Accounts of coextensive space highlight how digital and physical layers blur in XR [57], while social exchange frameworks link avatars, objects, and cues to trust and reciprocity [58].

Work on social presence and collaboration provides rich insight into how gaze, embodiment, and multimodality shape shared experience [1–3, 42, 55]. Yet these studies rarely consider kitchens as materially dense and socially charged environments. Our study brings these insights into connected kitchens by analysing how speculative phygital artifacts shape presence, coordination, and cohesion within domestic routines.

2.4 Inclusivity, Accessibility, and Ethical Considerations in Connected Domestic XR

Inclusive design plays a growing role in domestic XR and smart home research. MR and VR-based AAL systems support older adults and users with mobility or cognitive challenges [39, 59]. Digital twin–MR collaborations help dispersed families assist elders remotely [60]. Social VR exergames and virtual cultural experiences for older adults show how shared embodied tasks strengthen connection for specific user groups [61, 62]. Mixed Dining foregrounds social interaction among elderly individuals eating alone [21]. Domestic XR has also been used to explore sensory difference and family dynamics. *Cooking in the Dark* embodies blindness to foreground sensory-motor accessibility challenges [19]. ADHD-oriented co-regulation displays [9] and negotiation-oriented ambient systems like *Schlusslicht* [29] demonstrate how domestic technologies scaffold regulation and shared awareness. Passive co-presence research highlights how ambient sound, tactile cues, and object traces sustain family connection across distance [10, 33].

Ethical concerns are emerging prominently. Gaze-based identification research reveals privacy risks in social VR [63]. Telepresence robot design emphasises balancing embodiment, control, and acceptability in intimate domestic settings [64]. Social exchange research reframes XR interactions through trust, agency, and long-term relational dynamics [58], while presence research warns against assuming presence is universally beneficial [56]. Domestic XR research increasingly attends to accessibility, care, and ethical risk [10, 29, 33, 59, 60, 63]. Yet kitchens remain under-explored as active domestic workspaces where inclusivity, privacy, and social communication intersect. Our work examines speculative phygital artifacts as relational and ethical actors in connected kitchens, highlighting how such systems may support—or complicate—equitable and sustainable forms of domestic social interaction.

3 Methods

We designed our methodological approach to understand how professional designers interpret speculative phygital artifacts and imagine social interaction in hybrid connected kitchens. To address our research questions, we employed the quantitative and qualitative methods. We conducted an immersive VR-based evaluation study followed by semi-structured interviews and a design-evaluation survey. The quantitative results are derived from the analysis from the survey as a control/validation mechanism of the qualitative results which are the results of analysis of outcomes from interview and open ended questions from survey.

3.1 Methodological Approach

Our study followed a research-through-design-inspired evaluation protocol, in which interactive speculative artifacts were embedded into an immersive VR kitchen environment and experienced by participants through situated tasks. This combination of experiential exploration, reflective interviewing, and structured survey responses allowed us to investigate how designers make sense of synchronous and asynchronous communication concepts and how they envision everyday practices emerging around them. The methodological structure was deliberately aligned with our research questions: the VR sessions offered first-person experiential grounding for RQ1 and RQ2, the interviews elicited deeper reflections on social presence, collaboration, and concerns, and the survey provided structured comparative insights across design, interaction, usability, and contextual fit.

3.2 Participants

Fourteen professionals from multiple departments at the headquarters of a global home appliance company in [blind] took part in the study. Participants were recruited using purposeful expert sampling through an internal open call targeting teams engaged in product development, innovation, and design. Sessions took place in a dedicated testing room within the company. Participants ranged in age from 26 to 45 (median: 38). All participants provided informed consent.

Table 1 Participant IDs, demographics, educational background, field of expertise, years of experience, and their self-reported familiarity with kitchen and home-appliance design and VR/MR experience level (7-point Likert scale).

ID	G	Age	Edu	Field	Exp	Prod. Design Exp.	VR-MR Exp.
P1	M	37	Bachelor	Product Design	11	5	5
P2	M	40	Master	Engineering	15	7	5
P3	M	45	Master	Product Design	15	7	4
P4	M	40	Bachelor	Product Design	14	7	4
P5	F	31	Master	Product Design	6	5	3
P6	M	41	Bachelor	Product Design	15	5	5
P7	F	34	PhD	HCI	6	7	3
P8	F	26	Bachelor	Product Design	3	7	1
P9	M	34	Master	HCI	10	7	7
P10	F	28	Bachelor	Product Design	4	5	4
P11	M	43	Master	Product Design	15	5	4
P12	F	39	Master	Engineering	15	4	4
P13	F	32	Bachelor	Product Design	10	5	5
P14	M	39	Master	Product Design	15	7	4

3.3 VR Experience Procedure

The study consisted of three sequential phases: a VR experience phase, a semi-structured interview, and a post-study survey. After a brief introduction to the purpose, equipment, and procedure, participants provided informed consent and were guided into the VR environment. Each participant interacted with three speculative communication artifacts using a think-aloud protocol, while the facilitator observed from the physical environment and simultaneously joined the VR scene through a multiplayer setup. The facilitator prompted participants to verbalize thoughts throughout the session as a part of the thinking aloud protocol. All sessions were recorded using screen capture and external audio/video.

VR Kitchen Environment

For this study, we developed a dedicated VR prototyping environment inspired by the structure of our prior tool (blinded for review), but rebuilt from the ground up to avoid identity disclosure. The environment featured a fully interactive U-shaped kitchen with functional cabinets, appliances, a sink, utensils, and a variety of food items. Scenes were modeled using Blender3D, implemented in Unity, and scripted using MetaXR SDK components combined with custom interaction logic. Multiplayer functionality was enabled through Photon Fusion and Meta Avatar SDK. This environment allowed us to embed the three artifacts, Ambient Cooking, Portal, and Magmemo, into a coherent spatial and interaction-rich domestic setting.

Artifact Selection Rationale

The three artifacts examined in this study, Ambient Cooking, Portal, and MagMemo, were selected through an exploratory concept development phase in which we elaborated on common communication challenges in domestic kitchen environments on cooking, remote togetherness, and ambient domestic technologies. From a broader set of early concepts, we intentionally chose three artifacts that represent complementary modalities of hybrid communication: (1) an *ambient*: *Ambient Cooking* modality that foregrounds mood, atmosphere, and low-demand co-presence; (2) a *synchronous*: *Portal* modality that supports high-bandwidth, real-time social interaction; and (3) an *asynchronous*: *MagMemo* modality that enables expressive, persistent messaging embedded in the physical environment. This trio allowed us to probe a diverse spectrum of spatial and temporal communication practices while remaining feasible to prototype within a shared VR kitchen environment.



Fig. 1 The VR kitchen environment used in the study. Participants interacted with a fully functional U-shaped kitchen featuring food items, appliances, and spatially embedded prototypes.

Ambient Cooking

In the first stage, participants engaged in an interactive pizza-garnishing task designed to foreground ambient, atmospheric qualities. The kitchen lighting was adjusted to evening conditions, and participants used bowls containing ingredients such as mozzarella, olives, peppers, corn, ham, sausage, and pepperoni. Ingredients were poured or placed onto dough via shaking or grasping interactions; the system instantiated corresponding ingredient meshes dynamically. As the task began, the Ambient Cooking prototype activated: an AI-generated animated visual of a Mediterranean seaside town (integrated as a subtle moving backdrop) and ambient soundscape played, complemented by soft colored countertop reflections. The goal was to simulate how ambient and aesthetic cues might shape social and emotional experiences during cooking.



Fig. 2 The default Portal design viewed by participants. The circular frame includes ambient blue backlighting and subtle visual distortion, offering life-size synchronous video and audio from another kitchen space.

The Portal

The second stage introduced the Portal—a synchronous communication artifact inspired by ancient architectural motifs and their popular reinterpretations in science-fiction media. The Portal enabled real-time audio-visual communication and provided a parallax effect, allowing participants to shift viewing angles through embodied movement [65, 66]. To simulate communication across two distinct households, a second kitchen was added to the scene. The facilitator moved into this secondary kitchen in VR, leaving the participant alone to explore portal behaviors. Participants were then guided through a scenario where they described their previous pizza creation using hand gestures and explored how portals might support everyday communication. Portals were implemented using the *Portals for VR* Unity asset (limited to video and audio transmission). Multiple form-factor variations—including circular, compact, square, and rectangular portals—could be toggled by the participant.

MagMemo

In the final stage, participants explored MagMemo, an asynchronous communication artifact inspired by refrigerator magnets used for notes and reminders in everyday homes. Six 3D+audio messages were recorded by designers (and in one case, a child), each presenting a distinct domestic scenario such as forgotten items, grocery requests, recipe sharing, or dinner invitations. Messages were captured using depth-enabled mobile-phone cameras and reconstructed as animated 3D mesh sequences. In VR, each MagMemo appeared as a framed object pinned to the refrigerator; participants activated them using point-and-click interaction to trigger playback.



Fig. 3 A participant (shown as a Meta Avatar) interacting with Magmemo. Each message is represented as a framed 3D animation with accompanying audio, activated through point-and-click interaction.

3.4 Semi-Structured Interview

Following the VR experience, participants took part in a semi-structured interview consisting of three parts. First, they were asked about VR sickness symptoms and general impressions of the VR environment. Second, for each artifact, three guiding questions prompted evaluative reflection, imagined use cases, and broader design considerations. Finally, participants identified which artifact they felt held the most value for further development and expressed any concerns—including privacy, usability, and ethical issues. Each interview held one by one and recorded in audio, which are then transcribed from the analysis.

3.5 Survey

To complement the qualitative insights, a survey was administered including 18 design-evaluation items per artifact and six demographic questions. Sixteen items of artifact questions used a 7-point Likert scale; two were open-ended. Items were grouped into four constructs—Design, Interaction, Usability, and Contextual Fit—allowing comparison across conceptual dimensions and across artifacts.

3.6 Data Analysis

In our analysis, we integrated a qualitative thematic examination of notes taken during the VR exploration, interview responses and open-ended survey items with a quantitative evaluation of usability metrics. This mixed-method approach strengthened the validity of the designers’ feedback and provided more comprehensive insights into the overall experience and the proposed artifacts. In qualitative analysis we followed the six-phase reflexive thematic analysis framework by Braun and Clarke [67], an approach well suited to exploratory studies seeking to understand user experiences and professional practices. Two researchers independently reviewed the dataset, documenting observations and identifying preliminary patterns. Using an inductive coding strategy,

No.	Statement	Category
1	The Artifact is well designed in outlook (form, color, material, size, etc).	Design
2	The Artifact is well designed in terms of interactions.	Interaction
3	The Artifact needs a redesign in terms of usability, security, privacy, or other aspects.	Design
4	The Artifact uses intuitive interactions.	Interaction
5	The Artifact needs a redesign for better interaction.	Interaction
6	The Artifact should provide more ways to interact.	Interaction
7	The Artifact is designed flexibly and provides sufficient control to users.	Design
8	The Artifact fits well into the kitchen context.	Context
9	The Artifact is well positioned in the room.	Context
10	The Artifact is user-friendly and accessible to most users.	Usability
11	The structure, technology, and techniques of The Artifact are clear to understand and imagine.	Usability
12	The Artifact has potential for future contexts such as cooking, communication, smart home, or others.	Context
13	The Artifact has user potential in today’s context.	Context
14	There are competitive tools/technologies available today that can replace The Artifact.	Context
15	The Artifact fits well into domestic environments outside the kitchen.	Context
16	The Artifact fits better into non-domestic spaces (public, work, meeting rooms, etc.).	Context

Table 2 The 7-point Likert survey items used to evaluate each artifact, categorized into Design, Interaction, Usability, and Contextual Fit.

we allowed codes to emerge directly from the data. These codes reflected participants’ reflections on social presence, collaboration, and concerns.

4 Findings

This section presents the results of our mixed-methods study. The survey data provide a structured evaluation of each artifact’s design, interaction qualities, usability, and contextual fit, while the interviews and think-aloud observations offer deeper insight into how designers interpreted the prototypes and imagined their place in everyday kitchen life. Together, these complementary perspectives reveal both how the artifacts were assessed and how they were envisioned as components of future hybrid connected kitchens.

4.1 Survey Findings: Artifact Evaluation

Across all three prototypes, participants rated contextual fit in the kitchen and perceived future potential particularly positively, indicating that the overall concept of hybrid connected kitchen artifacts was well received. Portal and MagMemo tended to receive more consistently positive scores on appearance and interaction, whereas the Ambient Cooking Experience elicited more polarized ratings, especially around interaction intuitiveness and redesign needs. At the same time, all three artifacts were seen as candidates for further refinement, with respondents frequently highlighting the need for richer interaction methods and improved usability, privacy, or safety.

4.1.1 Portal

Overall, participants evaluated the Portal positively across design and contextual factors. Appearance (Median = 5, SD $\approx$ 1.26) and interaction design (Median = 5, SD $\approx$ 1.52) received consistently high ratings, and participants agreed that interactions were intuitive (Median = 5, SD $\approx$ 1.56). Contextual integration within the kitchen was among the strongest results (Median = 6, SD $\approx$ 1.12), alongside clarity of structure and technology (Median = 6, SD $\approx$ 1.41). The highest-rated dimension was perceived future potential across cooking, communication, and smart-home scenarios (Median = 7, SD $\approx$ 0.96).

At the same time, several items highlighted clear areas for refinement. Participants indicated a strong need for additional interaction methods (Median = 6, SD $\approx$ 1.50) and moderate agreement regarding redesign in terms of usability, privacy, or interaction improvements (Median = 5, SD $\approx$ 1.58). While competitive technologies were acknowledged (Median = 4, SD $\approx$ 1.70), participants still viewed the artifact as adaptable beyond the kitchen into broader domestic and, to a lesser extent, public contexts.

4.1.2 MagMemo

Participants also evaluated MagMemo favorably across several core dimensions. Appearance (Median = 4, SD $\approx$ 1.15) and interaction design (Median = 5, SD $\approx$ 1.10) were rated positively, and participants generally agreed that the artifact enables intuitive interactions (Median = 5, SD $\approx$ 1.60). Contextual fit in the kitchen showed consistently strong endorsement (Median = 6, SD $\approx$ 1.18), and the clarity of the artifact’s structure and technology was rated above the midpoint (Median = 6, SD $\approx$ 1.39). As with Portal, a strong signal emerged around future potential, with high agreement regarding its suitability for cooking, communication, or smart-home scenarios (Median = 6, SD $\approx$ 1.00) and solid perceived usefulness in current contexts (Median = 5, SD $\approx$ 1.24).

MagMemo likewise revealed room for improvement. The need for additional interaction methods showed strong agreement (Median = 6, SD $\approx$ 1.67), and moderate support emerged for redesign related to usability, privacy, or safety (Median = 5, SD $\approx$ 1.41). Some respondents acknowledged the presence of competitive technologies (Median = 5, SD $\approx$ 1.71). Adaptability beyond the kitchen received generally positive but more variable ratings (domestic non-kitchen spaces: Median = 6, SD $\approx$ 1.46; out-of-home contexts: Median = 5, SD $\approx$ 1.62), suggesting that participants viewed the concept as extendable but increasingly context-dependent outside domestic environments.

4.1.3 Ambient Cooking Experience

Evaluations of the Ambient Cooking Experience were more polarized than for the other two artifacts. Appearance (Median = 5, SD $\approx$ 1.33) and interaction design (Median = 5, SD $\approx$ 1.86) received moderate-to-positive ratings, yet intuitive interaction was endorsed less consistently (Median = 5, SD $\approx$ 2.12), indicating greater variability in perceived ease of use. Contextual fit nonetheless remained strong overall: participants

agreed that the artifact integrates well within the kitchen (Median = 6, SD $\approx$ 1.44), is appropriately positioned (Median = 6, SD $\approx$ 1.60), and is generally understandable in terms of structure and technology (Median = 6, SD $\approx$ 1.67). Future potential again emerged as a highly rated dimension, with strong scores for suitability across cooking, communication, and smart-home scenarios (Median = 6, SD $\approx$ 1.33), as well as for current user applicability (Median = 6, SD $\approx$ 1.15).

At the same time, critical perspectives were more pronounced for this artifact. Participants gave more mixed responses regarding the need for redesign in usability, privacy, and safety (Median = 3, SD $\approx$ 2.13), suggesting that some saw a clearer need for change than others. The need for additional interaction methods was strongly endorsed (Median = 6, SD $\approx$ 1.86), and competitive technologies were acknowledged at similar or higher levels than for the other artifacts (Median = 5, SD $\approx$ 1.41). Adaptability outside the kitchen showed mixed support: while participants considered the artifact suitable for other home contexts (Median = 6, SD $\approx$ 1.33), ratings for applicability in public or professional settings exhibited wide variance (Median = 5, SD $\approx$ 2.10). Overall, the results suggest that although the Ambient Cooking Experience is viewed as contextually meaningful and future-ready, it raises clearer concerns around interaction adequacy and redesign requirements.

4.2 Interview and Observation Findings: Designer Reflections

Designers' reflections, drawn from think-aloud sessions, interviews, and open-ended survey responses, converged around four recurring areas that shaped how they imagined these speculative artifacts becoming part of everyday kitchen life. The themes of Design and Flexibility, Privacy and Security, and Cooperation and Communication capture how participants envisioned the artifacts adapting to different routines, protecting the intimacy of the home, and enabling new forms of synchronous and asynchronous social connection. Complementing these, the Use Cases theme highlights the alternative scenarios designers proposed.

4.2.1 Design & Flexibility

For the interviews, we prepared a set of prompting questions aimed at sparking discussion around design decisions such as form, size, interaction, placement, and flexibility. These questions were asked separately for each artifact. In addition, each participant was asked how they would design the related artifact or what kinds of changes they would make. Flexibility emerged as one of the shared and significant outcomes across participants. All participants emphasized flexibility in the design; under this concept, they referred primarily to the mobility and personalization of the artifacts. This may be related to the different placement options of the Portal that participants tried during the experience.

"In its current design, the portal feels like a kind of video-chat tool. If I were to design it, the first thing I would do is free the portal from fixed forms and fixed positions. In my opinion, if the portal were something that the user could create in any position and in any form using the controller, the potential interactions would increase significantly. For example, in the kitchen context, the user could place the portal on the countertop or under

the upper cabinets if needed. This would allow the interactions enabled by the technology to expand.” (P7)

Feedback for the Portal included comments in a similar direction, such as: *“I would prefer it to be flexible in terms of placement—being able to move it to a different location” (P10)*, and *“... it could even have a feature that follows me as I move around the house.” (P1)*. Comparable observations appeared in interviews: *“Sometimes I cook here, then I move to the table. The portal should come along too.” (P7, P13)*. Participant 7 also suggested more imaginative options: *“I would like to be able to draw/place that portal wherever I want... like with a laser or something.”* While these quotes clearly point to mobility, some participants suggested partial flexibility for the Portal: *“Being able to adjust the portal’s position depending on where the person spends most of their time... there should be 2–3 placement options.” (P4)*. Beyond placement, others highlighted interaction flexibility: *“...I might even want to move the camera around—to show what’s on the stove, inside the oven, or something on the countertop.” (P13)*

Insights about the Ambient Cooking Experience focused more on personalization. During the interview, P6 noted: *“If I can choose the music or the ambiance myself... that would be great. It helps with motivation.”* Similar opinions appeared in the survey: *“It would be good if the content were under my control whenever I want, or if I could use AI.”* and *“Changing the music, increasing/decreasing the volume, changing the light color (colors adjusted according to context and time of day, or selected from predefined favorites).”*

Flexibility suggestions were less frequent for the MagMemo artifact, even though some participants questioned its placement: *“...I should be able to see it wherever I want... otherwise it would make me question why I’m forced to look there.” (P3)*. Flexibility in design was articulated more concretely as: *“Being able to change the size. I would also want features such as start, stop, add/remove, hide, and volume on/off.” (survey)*

Beyond flexibility and personalization, participants also shared a range of additional design suggestions. One commonly recurring idea was expressed as follows: *“Instead of appearing like a new product in an already cluttered kitchen, it should have a built-in, concealed design.” (P4)*. This feedback highlights an important point about how designers conceive kitchen technologies—even virtual ones—as ideally integrated rather than additive. Another participant suggested reinforcing functionality and coverage by changing the form: *“Rather than being planar, the portal could have been a concave surface, creating a more immersive perception through its form.” (survey)*

Design suggestions for the Ambient Cooking Experience often focused on the content itself. Participants proposed ideas such as: *“If there were something that could create an ambiance like an actual Chinese kitchen... an American kitchen... or the kitchen of some famous cook in France, it would really put you in the mood.” (P4)*, and *“...it could include live views of certain cities or places.” (survey)*. One survey response also pointed to a perceived lack of interaction, noting: *“There was no interaction space for the user in this concept; this aspect could have been addressed more comprehensively.”* For the MagMemo artifact, most feedback emphasized noise and disruption in the 3D visual quality. Additionally, two different preferences for visual

representation emerged: *“Its bust-like appearance feels unpleasant to me... it might be better if it looked more integrated as a whole.”* and *“It doesn’t have to be a direct video recording. The avatars would already be loaded, so when the other person sends a written message, you could see their avatar and hear its voice.”*

4.2.2 Privacy & Security

Given the communicative nature of the artifacts, privacy emerged as a central concern for both participants and researchers. Several comments about the Portal shared similar worries; for example: *“In terms of security and privacy, it should not open directly; instead, it should have a structure that can be opened or closed only with the user’s approval.”* (survey), and *“I think the kitchen is a private space. I would use it to communicate with my spouse, friends, or mother, but you don’t always want it to be open at home. The person on the other side may not want to turn on their camera...”*. Three different comments on MagMemo proposed ways to address privacy. Two participants suggested access control mechanisms: *“Only my messages should be visible. Like Face ID.”* (P2), and *“The sender should be able to choose the visibility of the message.”* (P14). In contrast, P3 considered that, since these messages are sent to a household, they are inherently common messages intended for all occupants.

The integration of XR technologies into the kitchen also prompted designers to reflect on safety. Concerns about this issue concentrated largely on the Ambient Cooking Experience. One survey comment reflects this tension: *“I thought it might not help me focus better on what I am doing in the kitchen, but rather could be distracting. For me, good lighting and proper ergonomics are more important.”* In contrast, some participants suggested enhancing functionality to support cooking and safety, such as: *“It could warn me if the kettle is hot.”* (P6)

4.2.3 Cooperation & Communication

The communication artifacts, Portal and MagMemo, were frequently interpreted as tools for cooperation. In the communication dimension, the Portal was mostly associated with cooking and co-cooking scenarios, with feedback such as: *“You talk to your mother... and at the same time, you cook together with her.”* (P3) and *“It can be used for a synchronous cooking experience, allowing someone with limited cooking experience to learn by following a more experienced person.”* (survey). Additionally, this form of communication was imagined for assisting others, as expressed by P6: *“It could also be used for helping an elderly person.”* Further scenarios clustered around workshop-like uses, for example: *“A famous person giving a live cooking demonstration.”* (survey). Additional comments on cooperation also appeared under Section 4.2.4. Beyond cooking, lightweight social communication was discussed for both Portal and MagMemo by P6, P4, and P14, for example: *“Family far away... it’s good for easing that longing.”* (P6). P7 similarly described: *“My aunt once gave me a cornbread recipe... that paper has been with me for 8 years. It feels inherited—valuable—so I keep it. It would have been nice to record it in my aunt’s real voice, or even have a 3D model of the final bread, a 3D cue showing a critical step in the recipe, or maybe an image of her eating it or of the dinner table.”* (P7)

In line with this quote, 4/14 participants imagined MagMemo as an asynchronous communication tool for cooperation. Comments such as the following underline collaboration through annotation-like usage: *“I should even be able to create a shopping list; when it’s placed on the fridge, it can offer more interaction and usefulness, and it can be combined with different user needs.”* In the discussion of the Ambient Cooking Experience, the first participant imagined blending playfulness and cooperation, commenting: *“It could even offer a new, concept-driven cooking recipe experience... like a gamified family cooking activity.”* (P1)

4.2.4 Use Cases

The thematic analysis revealed various use cases from the survey and interview data. We divided these use cases into cooking and non-cooking categories. Although most cooking-related use cases appeared in earlier sections under the themes (Section 4.2), we also cover additional examples here. Non-cooking use cases include both in-kitchen and out-of-kitchen scenarios, as well as entertainment-oriented applications.

Cooking By the nature of the artifact context, all participants initially envisioned the use cases of Portal and Ambient Cooking within cooking scenarios. Additional examples were also mentioned, such as: *“It could be used to make the environment more enjoyable in amateur cooking classes.”* (P11) As discussed under cooperation, adding playfulness to cooking was also mentioned for both Ambient Cooking and MagMemo, with comments such as: *“It could even go into gamification—you’re chopping something and it guides you by projecting the cutting lines, it augments the sound of cutting, and it could even show how much 300 grams of cheese would be by projecting a guideline.”* (P14)

Non-Cooking Similar to P7’s comment about using the Portal artifact during dinner, P11 shared that *“A more holistic experience could have been combined with a kitchen scenario. It could be integrated into the meal-preparation process.”* Most participants (10/14) also agreed that the Portal has the potential to enhance communication between people beyond the kitchen, as expressed in comments such as: *“It could also be used while exercising or playing games.”* (P7). Similar ideas were shared for the Ambient Cooking artifact, including comments such as: *“It would be a technology that could be integrated into everyday life—it could be used for anything, even making cleaning a more enjoyable experience.”* (P14) Although some participants questioned anchoring MagMemos to the fridge, one survey comment supported this decision: *“Considering that the kitchen is the shared hub of the home, scenarios such as household members leaving notes for one another come to mind.”* Beyond communication and annotation use, participants also commented that MagMemo has strong potential as an augmented picture frame: *“I would want to collect content here related to memories I’d like to keep. That way, I could also place it in different locations around the house.”* (P7) and *“On the fridge... I would put the ‘We love you very much’ message my mom and sister recorded. I’d look at it whenever I feel like it.”* (P8)

5 Discussion

This study examines how design practitioners conceptually frame communication, collaboration, and social presence when kitchens become hybrid connected environments. The following sections discuss our insights across five thematic areas that respond to our research questions and situate our findings within broader scholarship on social XR, domestic HCI, and phygital interaction design. Each theme reflects a conceptual lens shaped by existing literature and the design space of domestic social technologies, while also pointing toward implications for future design research.

5.1 Reframing the Kitchen as a Hybrid Connected Space

Kitchens have long been recognized as socially charged domestic environments where coordination, care work, and interpersonal routines unfold. Prior HCI research on cooking and dining technologies has largely emphasized guidance, safety, and efficiency [6, 15], with some work beginning to foreground co-cooking and remote togetherness [4, 5]. Our findings extend this trajectory by framing the kitchen not only as a task-oriented workspace but as a *hybrid connected space* that can function as a social XR hub. Designer feedback highlights two intertwined aspects of this reframing. First, participants resisted fixed spatial configurations for XR artifacts in kitchens. As reflected in Section 4.2.1, the Portal was repeatedly described as something that should “come along too” when moving between preparation areas and the dining table, or be drawable “wherever I want” on demand. This insistence on mobility and reconfigurability suggests that hybrid connected kitchens should be conceived less as static installations and more as *reconfigurable communication zones* that flex with domestic rhythms. Second, designers framed the kitchen as a broader household hub. Under the non-cooking use cases (Section 4.2.4), participants imagined the artifacts supporting activities such as exercising, playing games, cleaning, or simply spending time together, aligning with domestic AR findings that highlight “social activity” and shared awareness as key values of home AR systems [11].

These insights suggest several implications for future design research. First, connected kitchen systems should explicitly support *spatial fluidity*: portals and ambient surfaces that can be repositioned, resized, or re-anchored across countertops, cabinets, and adjacent spaces rather than being bound to a single wall or appliance. Second, designers and researchers should treat the kitchen as part of a larger domestic network, designing for cross-room and cross-activity continuity instead of isolated cooking scenarios. Finally, the strong desire for concealed or built-in forms (e.g., designs that do not appear as “a new product in an already cluttered kitchen”) points to an agenda of *architectural integration* in which social XR functions are blended into cabinetry, lighting, and surfaces rather than introduced as standalone gadgets.

5.2 Social Presence and Co-Presence in Domestic XR Contexts

Social presence research typically focuses on work, play, or entertainment contexts, emphasizing immersion, non-verbal cues, embodiment, and shared attention as core drivers of presence and collaboration [1–3]. Our study brings these concerns into a domestic setting and shows that in kitchens, presence is evaluated not only in terms of

intensity or realism but also in relation to privacy, attentional demand, and household rhythms.

The three artifacts embody distinct modalities of presence. The Portal offers high-bandwidth synchronous presence through life-size audio-visual communication; Ambient Cooking foregrounds low-demand, atmospheric co-presence; and MagMemo supports asynchronous, memory-like presence via 3D+audio messages. Designers recognized the value of these different modalities but were cautious about when and how each should be invoked. For example, the Portal was associated with rich co-cooking and learning scenarios (e.g., cooking together with a parent, or receiving step-by-step guidance from a more experienced cook), yet participants also worried about having such a strong visual presence open “all the time” in a private kitchen. Similarly, Ambient Cooking was praised for its motivational, mood-setting qualities but flagged as potentially distracting or even unsafe if it drew attention away from heat sources or cutting tasks. These tensions echo broader findings that more presence is not always better [56] and that domestic technologies must adapt to complex temporal and social rhythms. Our results suggest treating presence in hybrid kitchens as a *tunable gradient* rather than a fixed design goal: households may need to shift between intense, synchronous presence (e.g., co-cooking, workshops), gentle ambient accompaniment (e.g., shared music and atmosphere), and lightweight asynchronous traces (e.g., messages, reminders) depending on time of day, task, and emotional state. Design research might therefore explore interaction mechanisms and policies that allow dynamic modulation of presence intensity, including “do-not-disturb” modes, scheduled or context-aware activations, and subtle transitions between presence modalities.

Moreover, the emphasis on supporting users’ competence and autonomy resonates with work on smart kitchen assistants such as “Cookie” [8], which balance proactivity with human agency. Future domestic XR systems should similarly ensure that social presence augments users’ sense of control and confidence rather than overwhelming them with constant connectivity or performative pressure.

5.3 Phygital Artifacts as Communicative Media in Everyday Life

Phygital systems—hybrid combinations of physical form, spatial interaction, and digital content—have been explored in smart homes, tangible interfaces, and ambient displays as expressive media for coordination, reminders, and emotional connection [10, 25, 28]. Our three artifacts can be read as such phygital communicative media: they are not simply tools for task completion but *spatial communication channels* embedded in the kitchen.

Designer reflections in Section 4.2.3 show how participants imagined these artifacts as supports for both everyday coordination (e.g., shopping lists, reminders, safety warnings) and more affective or playful exchanges (e.g., a cherished recipe recorded in a loved one’s voice, gamified family cooking activities, or ambient scenes that transport the household to another place). In these scenarios, the kitchen becomes a site where communication is enacted through light, sound, spatial framing, and embodied gestures—not just through text or voice. MagMemo, in particular, highlights the potential of phygital artifacts as *memory surfaces*. Participants envisioned it as an augmented

picture frame or a repository for important family messages that could be revisited over time, aligning with prior work on passive co-presence and situated traces in domestic environments [33]. At the same time, the critiques of its visual form (e.g., discomfort with “bust-like” representations, or a desire for more stylized or avatar-based renderings) underscore the importance of carefully chosen representational strategies when embedding personal content into shared spaces.

For future design research, these insights point to at least three directions. First, there is value in designing *families of phygital artifacts* that together span synchronous, ambient, and asynchronous communication, allowing households to compose their own communication ecologies rather than relying on a single monolithic system. Second, materiality and integration matter: artifacts that appear as lightweight additions or overlays may be perceived differently from those that are structurally embedded into furniture, walls, or appliances. Third, the expressive range of phygital artifacts could be expanded to include additional modalities (e.g., haptics, scent, or subtle motion), building on prior multi-sensory work in domestic care contexts [32] while remaining sensitive to clutter and overload.

5.4 Tensions Around Privacy, Boundaries, and Domestic Inclusivity

Domestic technologies inevitably raise questions around privacy, agency, inclusivity, and the negotiation of shared space. Prior research on smart homes, telepresence, and social VR has highlighted concerns about surveillance, control, and the long-term implications of always-on sensing [60, 63, 64]. Our findings show how these concerns play out specifically in the context of hybrid connected kitchens.

Participants articulated clear expectations for *boundary mechanisms*. For the Portal, they emphasized that it “should not open directly” and should only be activated with explicit user approval, reflecting a desire to avoid accidental exposure in a private setting such as the kitchen. For MagMemo, designers proposed granular visibility controls (e.g., FaceID-style access or sender-defined visibility), while others argued that messages posted in a shared kitchen should be treated as inherently collective. These perspectives echo ongoing debates in domestic HCI about shared versus individual ownership of data and displays in multi-person households. Concerns about safety and cognitive load surfaced most strongly around the Ambient Cooking artifact, where some participants worried that immersive ambience might distract from core cooking tasks and undermine ergonomics. At the same time, designers suggested safety-enhancing extensions such as warnings about hot kettles or other hazards, indicating that hybrid kitchen systems must carefully balance ambience and assistance.

Inclusivity emerged more implicitly, through comments about placement flexibility, clutter, and integration into existing kitchen layouts. The desire to reposition artifacts and to avoid additional “products” in already crowded workspaces suggests that future connected kitchen systems must account for diverse body sizes, mobility constraints, and sensory sensitivities. These concerns resonate with work on ambient displays for families with ADHD children [9], Ambient Assisted Living systems [39], and negotiation-oriented domestic artifacts such as Schlusslicht [29], all of which highlight the need for configurable, transparent, and respectful domestic technologies.

For design research, this points to a clear agenda: hybrid connected kitchens should be designed with *privacy-by-default*, explicit opt-in mechanisms for high-bandwidth communication, fine-grained message visibility and retention settings, and accessible controls for muting, relocating, or reshaping artifacts. Future studies might investigate how these controls are negotiated within households over time, and how hybrid kitchen systems can support vulnerable or marginalized household members (e.g., children, elders, or people with sensory differences) without reinforcing existing power imbalances.

5.5 Methodological Reflections: Speculative VR Probes for Domestic Futures

Our methodological approach combines speculative artifact design with an immersive VR prototyping environment, extending prior work on hybrid smart homes [27], AR-in-VR simulation platforms [13], and VR-based kitchen tools such as KitchXR [14]. By embedding Ambient Cooking, Portal, and MagMemo in a coherent, interactive VR kitchen, we were able to give designers a first-person experience of not-yet-available domestic technologies and prompt situated reflection on their communicative, emotional, and ethical implications.

Participants used this environment to experiment with placement, scale, and interaction styles, and to imagine use cases that extended beyond the scripted tasks. The think-aloud protocol and subsequent interviews revealed how speculative VR probes can surface both enthusiasm and skepticism: designers could articulate nuanced scenarios of togetherness, play, and support, while also identifying privacy concerns, aesthetic misalignments, and potential safety issues. This aligns with broader arguments in design research that speculative and immersive methods are well suited to probing emerging socio-technical configurations and making tensions more tangible.

At the same time, our approach has limitations. As a lab-based, time-bounded study with single participants, it cannot capture long-term appropriation, the dynamics of multi-person households, or the material constraints of real kitchens. The prototypes, though interactive, were not fully functional networked systems; this may have shaped participants' expectations and the granularity of their critiques. Future methodological work could therefore combine VR-based speculative probes with longitudinal in-home deployments of lower-fidelity artifacts (e.g., screen-based proxies, tangible mock-ups, or AR overlays), as well as cross-cultural studies that explore how different domestic norms and housing configurations shape the reception of hybrid kitchen technologies. For design research, a promising direction lies in developing *iterative hybrid pipelines* where concepts are first explored in VR (for spatial and experiential reasoning), then materialized as partial physical prototypes, and finally re-embedded into extended domestic trials. Such pipelines could also integrate participatory methods with non-designer household members, broadening the perspectives beyond professional designers while preserving the critical and imaginative potential of speculative VR.

6 Conclusion

This study explored how professional designers envision social interaction, collaboration, and communication in hybrid connected kitchens by experiencing three speculative phygital artifacts within an immersive VR domestic environment. Through situated VR tasks, reflective interviews, and structured design-evaluation surveys, we examined how designers interpret synchronous and asynchronous communication tools, how they imagine everyday practices emerging around them, and what opportunities and tensions they see when domestic spaces become XR-enabled communication hubs.

Our findings contribute to the growing body of VR and HCI scholarship on domestic XR, phygital communication, and socially meaningful hybrid environments. First, we introduce the notion of *hybrid connected kitchens* as a conceptual frame that positions kitchens not merely as task-oriented spaces but as socially charged, relational sites where ambient, synchronous, and asynchronous cues can meaningfully support togetherness. Second, we provide empirical insights into how design practitioners make sense of speculative phygital artifacts—highlighting perceptions of social presence, emotional comfort, contextual appropriateness, and concerns around privacy, accessibility, and household rhythms. Third, we demonstrate how immersive speculative prototyping can serve as an effective research probe for eliciting designer interpretations of domestic XR futures, offering methodological value for design-led research on emerging technologies.

This work also carries limitations. Our participant group consisted exclusively of designers from a single global home appliance company, which may limit the diversity of perspectives and restrict generalizability to other user groups, households, or cultural contexts. The VR prototypes were speculative and not functionally complete, which may have influenced how participants imagined long-term use. Furthermore, the study focused on designers rather than everyday household members; future work should expand to families, children, older adults, and multi-person households to better understand lived domestic dynamics.

Future research could explore long-term appropriation of hybrid kitchen artifacts in real homes through longitudinal deployments, examine cross-household interactions among family members with varied cultural and sensory practices, and investigate how AI-driven adaptation, multimodal cues, or shared embodiment can support richer domestic communication. Extending this work beyond kitchens to other domestic spaces—or to inter-home ecologies that link multiple rooms, households, and contexts—offers fertile ground for understanding how XR might shape new forms of togetherness, support, and coordination in everyday life. Ultimately, this study contributes an initial but substantive step toward imagining socially meaningful, inclusive, and ethically grounded XR experiences for domestic environments, pointing toward a broader vision of connected realities woven into the textures of everyday home life.

Declarations

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this study did not require formal ethical review or approval. **Consent to participate.** All participants were informed about the purpose and procedures of the study and provided written informed consent prior to participation. **Consent for publication.** Participants consented to the use of anonymized data and study outcomes in academic publications.

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